The Execution Boundary

How Much of a Measurable Short-Horizon Edge
Can Actually Be Realised, and What Determines the Limit

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1 The question

Short-horizon direction in a modern electronic limit order book is predictable to a degree that would be remarkable in any other asset class. In my own measurements on cryptocurrency perpetual futures — 236 features at 100 ms cadence, 2.17 million ticks per symbol on BTC, ETH and SOL — order-book imbalance attains a rank information coefficient of up to 0.47 at horizons of one to five seconds, uniformly across symbols and across volatility regimes.

Almost none of it can be earned. Conditioning on the states in which a passive (maker) order would actually have been filled *and* the signal was directionally correct collapses that coefficient to approximately 0.03. The collapse is not estimation noise: it replicates on all three symbols, and it has a clear mechanism, because the fill event is not exogenous to the prediction. You are filled when someone chooses to trade against you, and they choose to do so when they are better informed than you are.

The observation that turns this from a disappointment into a research programme is the third number. Conditioning on the presence of *any* fill — without requiring the direction to have been right — *raises* the coefficient to 0.52. Writing F for the event that a resting order is filled and D for the event that the signal was directionally correct,

$$\mathrm{IC} \;=\; 0.45, \qquad \mathrm{IC} \mid F \;=\; 0.52, \qquad \mathrm{IC} \mid (F \cap D) \;=\; 0.03. \tag{1}$$

Information is not absent in the states where trading happens — it is abundant there. What is absent is information in the states where trading happens *on our terms*. The gap between 0.03 and 0.52 is therefore not a measurement of how weak the signal is. It is a measurement of how much of it adverse selection appropriates, and it invites a question with a real answer:

> **What is the achievable frontier between 0.03 and 0.52, and what does its position depend on?**

That is the thesis I propose. It is falsifiable, it is quantitative, its lower and upper bounds are already measured, and progress on it is legible: any proposed mechanism either moves the conditional coefficient or it does not.

2 Why it is worth four years

It is a bound, not a strategy. The literature on order-flow imbalance establishes predictability (Cont, Kukanov and Stoikov, 2014, and the subsequent deep-LOB work). The literature on adverse selection establishes that market makers lose to informed flow (Glosten–Milgrom; Kyle, 1985). What is thin is the quantitative bridge: given a measured unconditional signal, how much survives a realistic execution channel, and which properties of the signal, the venue and the execution policy determine the survival rate. Equation (1) is a way of posing that question that a dataset can answer.

It generalises past its venue. Nothing in (1) is crypto-specific. The decomposition applies wherever a passive order can rest and a counterparty can choose to take it. Crypto perpetuals are a convenient laboratory — continuous trading, no fragmentation across lit/dark venues, published funding and oracle prices, and on at least one venue a fully public per-position liquidation structure — but the object of study is a property of limit order markets.

It has an honest negative baseline. I have already established, on my own data and against my own interest, that the naive answers fail. Five strategies the platform initially identified as profitable were subsequently refuted under a stricter cost model and a corrected evaluation harness. A signal-combination result was retired. In a pattern-discovery study, six kernels significant after false-discovery-rate correction in sample yielded none that survived out-of-sample, two of them reversing sign. Starting from a set of things known not to work is a better position than starting from a set of things not yet disproved.

3 Preliminary work

The proposal is not speculative: the instrument exists and has produced results, all of which are written up as working papers.

Paper	**Result relevant to this proposal**
Predictive structure of the perp LOB	The census behind (1): two empirically orthogonal channels (direction, volatility), eight independent directional axes, spectral localisation to 0.005–0.1 Hz, OU half-life 5–7 s, and the fill-conditioning decomposition.
Autonomous alpha discovery under FDR control	The apparatus: a five-gate replication protocol under Benjamini–Hochberg control with an adaptive threshold, and the first statement of the conditional-IC collapse.
The process as an analytical unit	The orthogonality and effective-breadth mathematics; why the binding correlation is the stress-regime one.
Event-aligned SVD	Pattern discovery with basis estimation provably blind to forward returns; a clean in-sample positive that does not replicate out of sample.
Exact or nothing (`nat2`)	A gate-governed measurement instrument in which lookahead is queryable by construction, plus a negative result on liquidation-map coverage.
Liquidation geometry; Prism; narrative as exogenous signal	Adjacent instruments and their stated, unvalidated protocols.

Supporting infrastructure: a Rust ingestion engine sustaining 100 ms feature emission with end-to-end latency under 80 ms at the 99th percentile; a research CLI of some 340 commands; a test suite of roughly 4,300 tests; and a hash-chained ledger in which a failed verdict cannot be quietly removed from the record.

4 Proposed work packages

WP1 — Characterise the boundary (year 1). Establish (1) as a robust, replicated fact rather than a single-window observation. Out-of-time and out-of-venue replication; decomposition of the collapse into queue position, cancellation dynamics, and counterparty composition;

and identification of which of the eight independent directional axes lose the most to conditioning. *Deliverable:* the measurement paper, extended and refereed.

WP2 — Model the fill as an endogenous, adversarial event (years 1–2). The standard treatment models fill probability as a function of queue position and volatility. That is insufficient here, because the relevant object is the joint law of (fill, signal correctness). I propose to model the fill indicator as adversarially coupled to the signal and to ask what execution policies are optimal under that coupling — a problem that admits both an optimal-stopping formulation (when to stop resting and cross) and a distributionally robust formulation (optimise against a worst-case fill measure within an ambiguity set around the empirical one).

WP3 — Mechanisms that move the frontier (years 2–3). Test, each with a pre-registered kill criterion: *(i)* regime gating — conditioning on book-shape entropy already recovers part of the loss in preliminary work; *(ii)* timescale separation — a slower signal harvested by patient execution should couple less strongly to the fill indicator, which is a testable prediction rather than a hope; *(iii)* orthogonal information sources, where the effective breadth argument $\mathrm{IR} \approx \overline{\mathrm{IC}}\sqrt{N_{\mathrm{eff}}}$ says a weak uncorrelated signal can beat a strong redundant one; *(iv)* the volatility channel, which is disjoint from the directional one and may be monetizable where direction is not.

WP4 — Synthesis (year 4). A characterisation of the frontier: given a signal's autocorrelation structure, its spectral support, and the venue's microstructure, how much of its unconditional information is realisable, and by which execution policy. Plus the negative counterpart — the conditions under which the answer is provably close to zero.

5 Data, method, and discipline

Data is in hand and accumulating: tick-level order book and trade data on three symbols at 100 ms, with a continuously running capture daemon, plus a position-level collector on a venue that publishes per-position liquidation prices.

The methodological commitments are already load-bearing in my own work and I would carry them into the doctorate. Every candidate result passes a replication protocol with false-discovery-rate control before it is believed. Every unit is tested against a planted synthetic signal *before* it touches real data — three estimator bugs were caught only that way. Lookahead is enforced by construction: each record carries both an event timestamp and an ingest timestamp, and a replay audit fails on any feature that reads what it could not have known. Verdicts, including failures, are appended to a hash-chained ledger, so the variants tried before the one that worked remain visible.

6 Risks, stated honestly

The frontier may be flat. It is possible that no execution policy meaningfully improves on 0.03 and that the answer to the thesis question is "essentially none of it, and here is why". That is a publishable result and the work packages are designed so that a negative answer is still a thesis, not a failure.

The venue may mature away from me. I have already measured efficiency drift in real time: the five-second signal weakened from 0.45 to 0.29 over six weeks as spreads tightened, while the one-second signal held. If the phenomenon disappears during the doctorate, its disappearance is itself a measurement of market efficiency arriving, and worth documenting.

Single-venue dependence. Mitigated by making the venue a parameter rather than an assumption in WP1's replication design.

7 Fit

What I bring is measurement infrastructure, a large clean dataset, and a demonstrated willingness to publish results that refute my own earlier claims. What I lack, and would come to learn, is the formal apparatus — optimal stopping, robust optimisation, high-dimensional inference — to turn a set of measured regularities into a theory of what is achievable. That asymmetry is the reason I am applying rather than continuing alone.